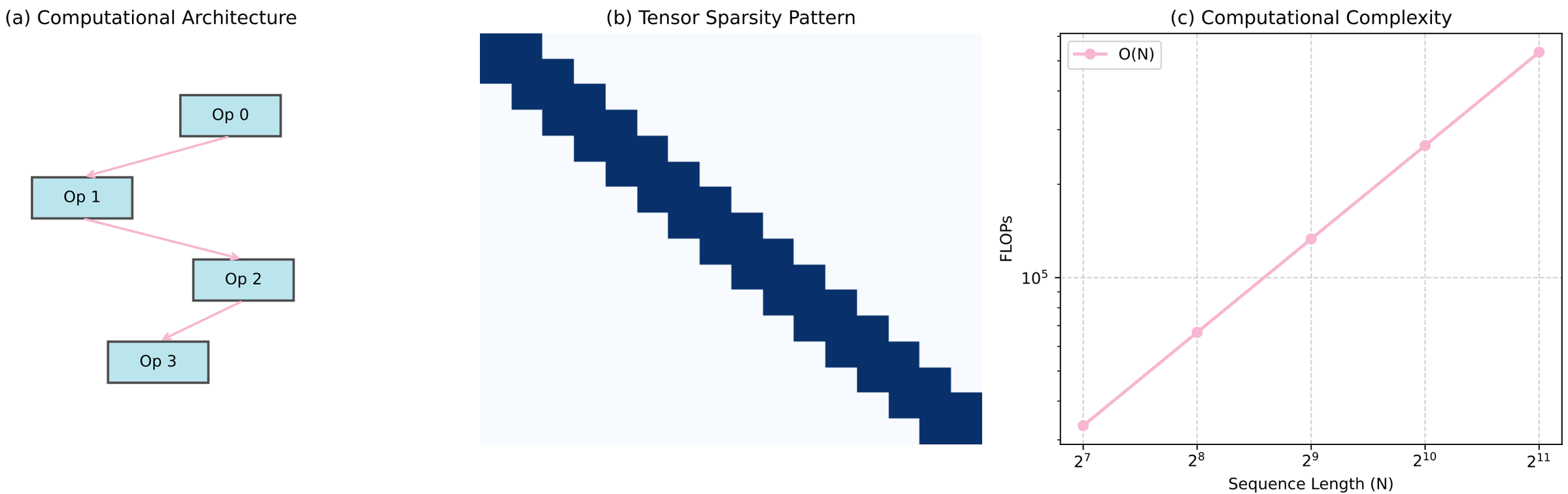Module Architecture: adaptive_compressor